

Project Name

Food Recognition And Calorie Estimator

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Introduction or Project Overview

Food recognition is an emerging application of computer vision with increasing importance in health monitoring, dietary management, and smart kitchen systems. Automatically identifying a food item from an image is a challenging task due to variations in lighting, plating style, background, and similarities between dishes.

This project focuses on developing a machine learning model that can **recognize 20 Indian food items from images** and additionally **predict approximate nutritional values** such as calories, carbohydrates, protein, and fats. Combining food recognition with nutritional estimation makes the system applicable to real-world use cases such as calorie tracking, diet logging apps, and personalized nutrition assistants.

The project uses deep learning, transfer learning, and data preprocessing techniques to build a robust food classifier. Nutritional values are estimated based on the predicted food class using a predefined nutrition lookup table.

Problem Statement

The main objective of this project is:

“To develop a machine learning model that identifies Indian food items from images and provides estimated nutritional values for the detected food.”

The problem presents the following challenges:

- **High intra-class variation:** The same dish may look different based on preparation style, angle, lighting, or serving method.
- **Inter-class similarity:** Visually similar dishes (e.g., curries or fried snacks) may confuse the model.
- **Limited dataset size:** Around 100 images per class increases risk of overfitting.
- **Nutrition estimation difficulty:** Nutritional values depend on portion size, ingredients, and preparation style—introducing uncertainty.

Despite these challenges, the system aims to deliver accurate classification and reasonable nutritional estimates for real-life usefulness.

Overview of the Dataset used

The project uses the **Food20 Dataset** from Kaggle, which contains:

- **20 Indian food categories**
- **Around 2,000 total images**
- **~100 images per class**
- High diversity in lighting, angles, and backgrounds

Dataset strengths:

- Balanced number of images
- Diverse appearance of dishes
- Focus on Indian cuisine, unlike many global datasets

Limitations:

- Moderate size
- Some foods with very similar visual features
- Images vary in resolution and background complexity

Project Workflow

○ Data Preparation

- Dataset downloaded and inspected.
- Train–validation–test split created.
- Data augmentation applied to reduce overfitting:
 - Rotation
 - Horizontal/vertical flips
 - Zoom & crop
 - Brightness/contrast adjustment

○ Model Architecture

- Transfer learning used with a pretrained CNN model :

EfficientNet-B2

- Final dense layers replaced with softmax layer for 20-class output.
- Batch normalization and dropout used to improve generalization.

○ Model Training

- Trained using:
 - Adam Optimizer
 - Categorical Cross-Entropy
 - Early Stopping
 - Learning Rate Scheduling

○ Nutritional Value Estimation Module

After the food is identified, a **nutrition lookup table** is used to return approximate nutritional values, such as:

- Calories (kcal)
- Protein (g)
- Carbohydrates (g)
- Fats (g)

This allows the system to function like a diet-tracking tool.

Example workflow:

1. User uploads an image
2. Model predicts: “*Biryani*”
3. Lookup table returns nutritional values for biryani
4. System displays predicted food + nutrition

○ Evaluation

- Accuracy measured on training, validation, and test sets
- Confusion matrix used to analyze class-level errors

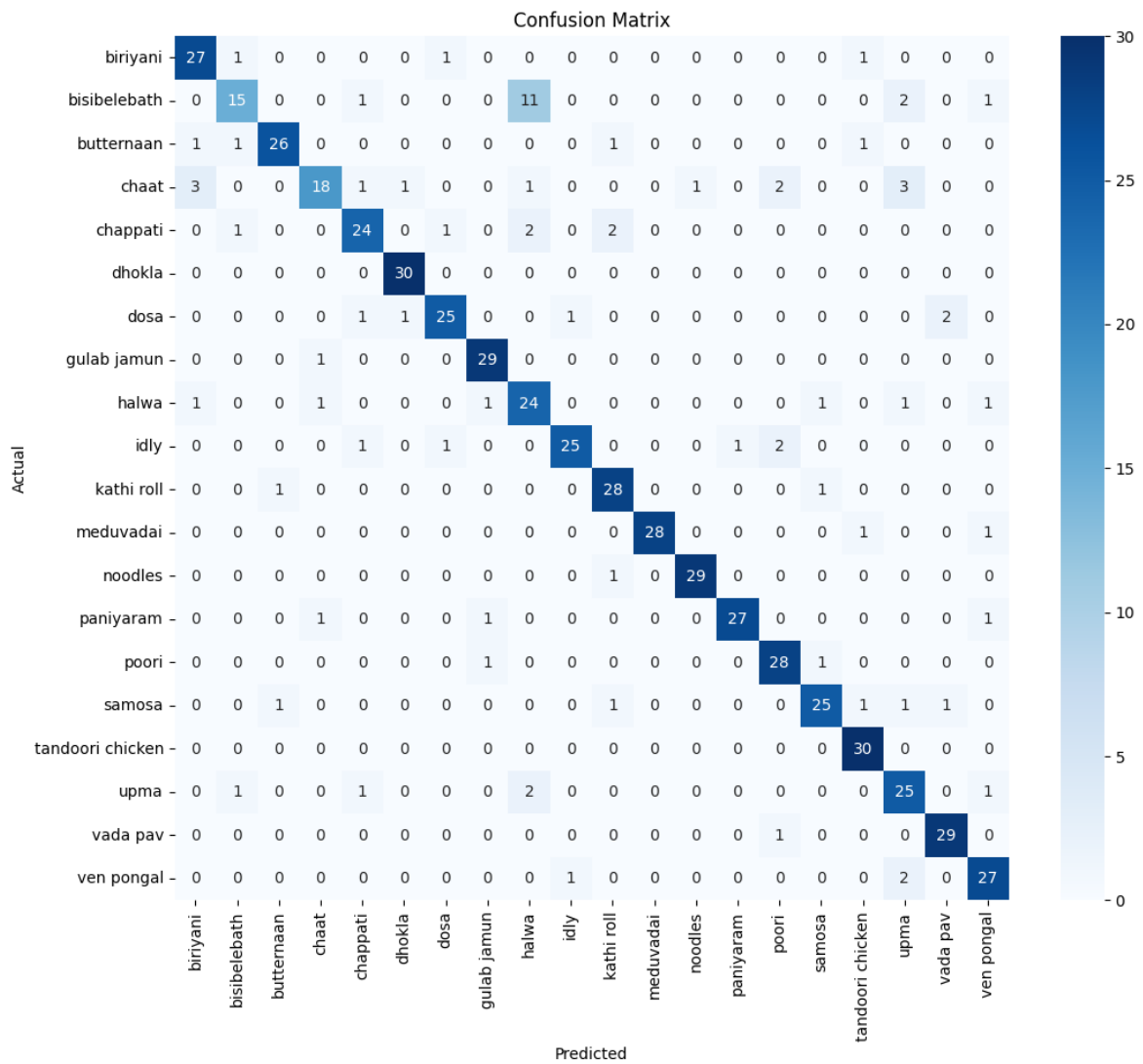
Results

○ Accuracy Scores :

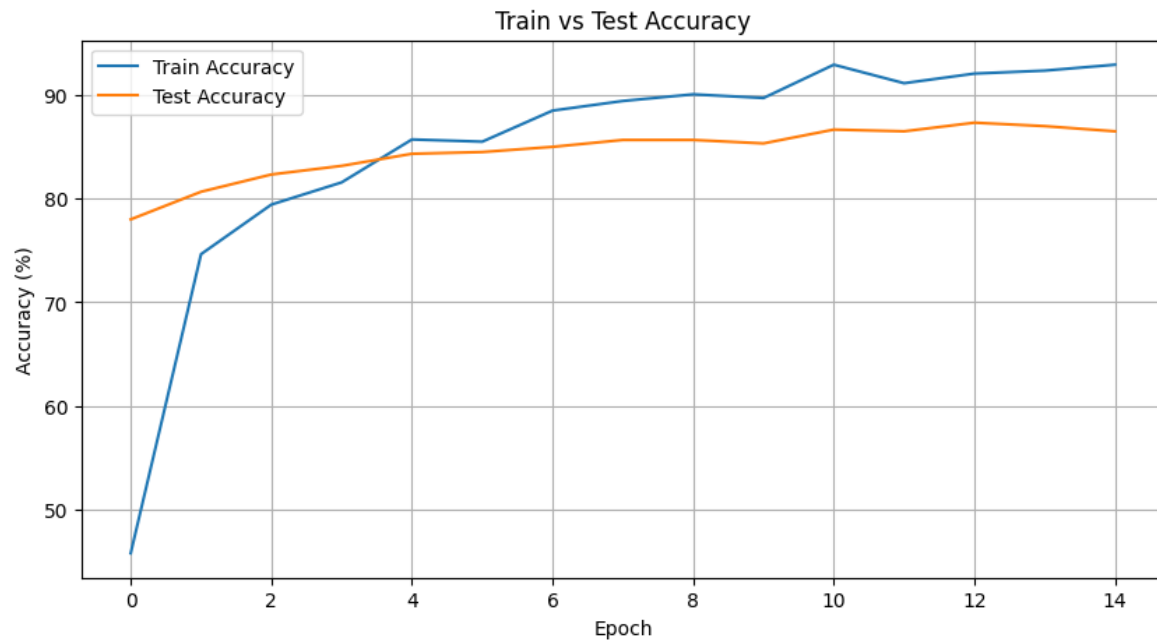
Training Accuracy – 92%

Test Accuracy – 87%

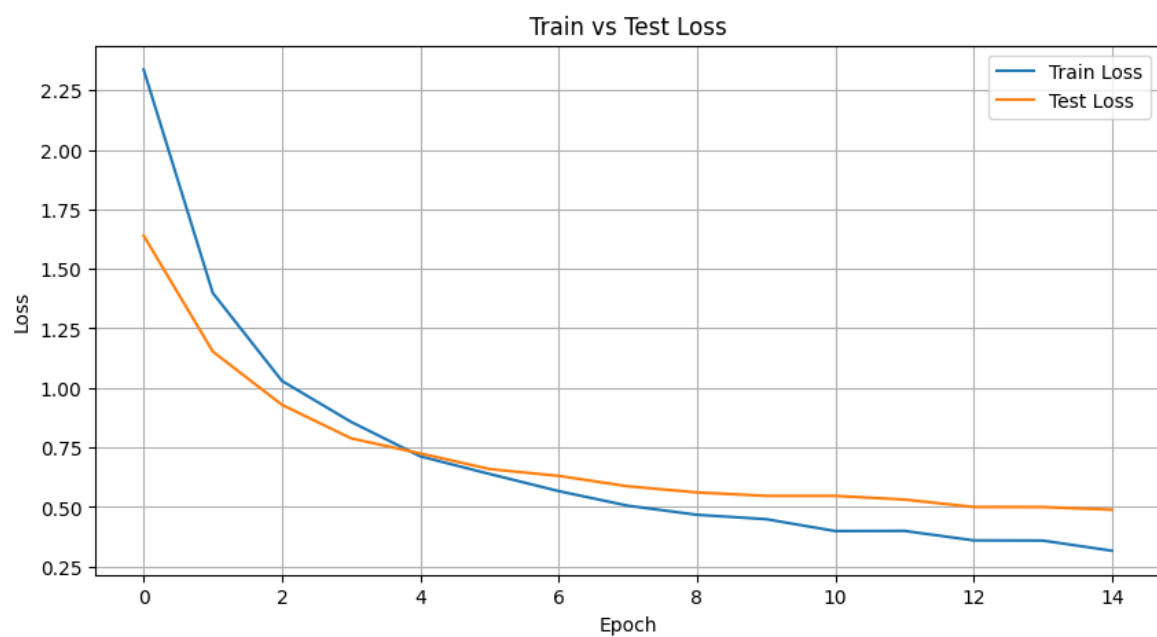
○ Confusion Matrix



○ Train vs Test Accuracy Graph



○ Train Vs Test Loss Graph :



Conclusion

This project successfully developed a food recognition system capable of classifying 20 Indian food items using deep learning. Transfer learning significantly improved accuracy, while data augmentation helped prevent overfitting. The addition of **nutritional value prediction** enhances the system's real-world applicability for health monitoring and dietary tracking.

Key Achievements

- Accurate classification of food images
- Integration of nutrition estimation module
- Strong overall performance with deep learning

Future Work

- Expand dataset to include more Indian dishes
- Predict portion sizes for more precise nutritional values
- Use Vision Transformers or EfficientNet for improved accuracy
- Deploy as mobile app with camera-based food recognition

The project demonstrates how machine learning can be applied to food computing and nutrition technology, opening doors to more advanced diet-monitoring tools.

